

MOHAK SHARMA

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SUMMARY

Product engineer and ML researcher with 3+ years shipping production systems. Founding Engineer at Subconscious AI, owning the hierarchical Bayesian statistical core, the multi-provider LLM layer, and two enterprise accounts, alongside an M.S. in Computer Science at Columbia and an Advanced Masters Research Award funding a post-graduation semester on EEG and gaze intent decoding. Bias toward research-rigorous systems that still ship.

EDUCATION

Columbia University, M.S. Computer Science, New York, NY

Expected Dec 2026

Coursework: Deep Learning, Causal Inference, Computational Neuroscience, Computer Vision I/II, Mechanistic Interpretability, Storage Systems.

Advanced Masters Research Award: Funded post-graduation research semester at LIINC on real-time EEG and gaze intent decoding integrated with embodied foundation models for human-in-the-loop robot control.

Guru Gobind Singh Indraprastha University, B.Tech. Computer Science, Delhi, India

2019 to 2023

EXPERIENCE

Subconscious AI, Founding Engineer, New York, NY

Apr 2026 to Present

- Sole technical point of contact for two enterprise accounts, Wargaming (350M+ registered players) and NRG (Fortune 500, \$30B+ revenue), delivering 20+ experiments and 5+ studies across a 500K+ respondent pool on a one-month turnaround, with a 12% conversion lift.
- Rebuilt the hierarchical Bayes conjoint estimator from scratch in NumPy (Gibbs sampling on the population mean and covariance, Metropolis-Hastings on respondent-level coefficients, constraints enforced in-sampler), layered per-respondent willingness-to-pay on it as a full distribution rather than a point estimate, and validated both against a 154-study social science replication corpus.
- Designed and shipped the Market Simulator, an interactive what-if engine for market share, revenue, and price sensitivity, with a Pareto frontier ranking re-priced scenarios by non-dominated sorting across three competing objectives, plus Concept Testing as a second experiment type.
- Owned platform infrastructure: multi-provider LLM integration across Bedrock, Grok, and Databricks, enterprise auth (Auth0 org SSO, role-based access), a unified network layer, and a proactively written golden-test suite covering the Pareto and WTP math.
- Built the agentic development infrastructure the team runs on: Sentry Seer and Datadog alerts auto-file ~40 triaged GitHub issues a week, Claude-driven agents draft the fix PR and its regression tests in Docker sandboxes, and quality guardrails gate every merge against functional regressions, holding 80% of agent PRs merged without rewrite and alert-to-merge at 6 hours.

Outscal, Forward Deployed Engineer, Growth and Product, Delhi, India

Nov 2023 to Jul 2025

- Architected an AI job-parsing pipeline processing 2,000+ postings daily (60K+ total), fine-tuning Gemini for structured extraction at 95%+ accuracy, and built a skill-matching recommender linking learning content to live job listings.
- Designed and ran A/B experiments across user journey touchpoints, analyzing cohort behavior and funnel metrics in PostHog, driving a 15% conversion lift.
- Rebuilt the web architecture on Next.js and GraphQL for 400K+ users, improving Web Vitals up to 800% down to ~100 ms on critical pages, and led an internal no-code CMS saving 60+ developer hours per week.

TravClan, Frontend Engineer, Delhi, India

Aug 2022 to Sep 2023

- Built a React internal operations panel for 250+ staff and booking features for portals used by 5,000+ agents; introduced a micro-frontend architecture that cut build sizes ~35% and raised deployment frequency 40%.

RESEARCH

LIINC Lab, Columbia University, Graduate Researcher

Mar 2026 to Present

- Designed and ran user studies on intent decoding: task and stimulus design, IRB protocol support, participant recruitment, and synchronized multi-stream capture of EEG, gaze, and behavior.
- Built the physiological inference pipeline end to end: EEG preprocessing (filtering, artifact rejection, epoching), gaze and pupil feature extraction, temporal alignment across asynchronous streams, and decoders mapping fused signal onto a latent intent representation, evaluated on cross-participant generalization.

CRIS Lab, Columbia University, Research Assistant

Sep 2025 to Jan 2026

- Probed representations across 48 transformer layers of AlphaFold2 over 200K+ AlphaFold DB structures using UMAP and contrastive analysis, relating Predicted Aligned Error structure to interaction interfaces via ChimeraX.

PROJECTS

- **CLARIS-Net, Neural-Driven Humanoid Control.** End-to-end BCI translating noisy EEG motor imagery into physics-grounded humanoid action under 100 ms latency; EEGNet with contrastive pretraining plus PPO/SAC refinement, reaching >90% accuracy on continuous joint-angle prediction for a 17-DOF agent in MuJoCo.
- **CausalBench.** Multi-agent evaluation harness that generates structural causal models and counterfactual tasks, benchmarking four frontier models (GPT-4, Claude, Gemini, Llama) on confounding, selection bias, and counterfactual reasoning, with accuracy landing between 40% and 65% on causal graph tasks.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL.

ML and Statistics: PyTorch, TensorFlow, NumPy, pandas, scikit-learn, transformers, CNNs, RNN/LSTM, contrastive and self-supervised learning, hierarchical Bayesian modeling (Gibbs, Metropolis-Hastings), causal inference, RL (PPO, SAC, DDPG), A/B testing.

Signal, Vision and Simulation: EEG decoding (MNE-Python, EEGNet, ICA, epoching), MuJoCo, PyBullet, real-time closed-loop control, POMDPs, NeRFs, COLMAP, Open3D, OpenCV.

Systems and Tooling: FastAPI, React, Next.js, Node.js, GraphQL, Zustand, MongoDB, Redis, AWS (EC2, S3, Lambda, CloudWatch, Bedrock), Databricks, Auth0, Linux, Git, PostHog.

LEADERSHIP

- **PDL Fellow, Columbia Engineering.** Selected by nomination into an invite-only cohort of under 35 graduate fellows.
- **Graduate Admissions Ambassador, Columbia University.** Outreach and mentoring reaching 2,000+ prospective graduate candidates annually.